Copernicus Global Land Operations
"Vegetation and Energy"
"CGLOPS-1"
Service Contract N° 945120-IPR-2023 (JRC)

VALIDATION REPORT
UPDATE 2024

SURFACE SOIL MOISTURE

VERSION 1.0

Issue I1.00

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Dissemination Level		
PU	Public	X
PP	Restricted to other programme participants (including the Commission Services)	
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Document Release Sheet

Book Captain	Bernhard Raml (TUWien)	Sign *Bernhard Raml*	Date 12.05.2025
Approval	Roselyne Lacaze (HYGEOS)	Sign *Roselyne Lacaze*	Date 05.06.2025
Endorsement	Nadine Gobron (JRC)	Sign	Date
Distribution	Public		

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Change Record

Date	Page(s)	Description of Change	Release
12.05.2025	All	First issue	I1.00

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TABLE OF CONTENTS

Executive Summary 12

1 Background of the document 14

1.1 Scope and Objectives 14

1.2 Content of the document 14

1.3 Related documents 14

1.3.1 Applicable documents 14
1.3.2 Input 15
1.3.3 Output 15
1.3.4 External Documents 15

2 Requirements 16

2.1 specific technical details and requirements 16

2.2 Further requirements 16

2.2.1 Output product composition 16
2.2.2 Data Structure 17
2.2.3 Data format 17
2.2.4 Uncertainties and Validation 17
2.2.5 Input data 18
2.2.6 Product delivery 18

3 Review of initial SSM1km Validations 19

4 Scientific Quality Monitoring Method 21

4.1 In-situ Reference Products 21

4.1.1 Used ISMN Networks 22
4.1.2 Spatial Representativeness 26

4.2 Validation procedure 27

4.2.1 Dataset Preparations 27
4.2.2 Assessment of the Temporal Signal Quality 28

5 Results 29

5.1 Overall Analysis Summary 29

5.1.1 Summary over Networks: Pearson Correlation Analyses 31
5.1.2 Summary over Networks: RMSD Analyses 35

5.2 Selected Time Series Analyses 39

6 Discussions and Conclusions 45
6.1 Summary from Dedicated Peer-Review Article 45
6.2 Quality Assessment over Europe 2015-2024 45
7 Recommendations and Perspectives 47
8 References 48

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List of Figures

Figure 1: Spatial distribution of ISMN stations used for the present SSM1km validation. 25

Figure 2: Box plot summarizing Pearson rho values for SSM calculated over all relevant ISMN stations (0-10 cm depth), per network. Data from 2015-2024 was included. The boxes indicate 25th and 75th percentiles, the whiskers are set to 5th and 95th percentile and outliers are indicated by individual points..... 31

Figure 3: Box plot summarizing Pearson rho values per network, based on data from 2015-2023 (brown) and 2024 (orange). Only networks with data in 2024 are included. The boxes indicate 25th and 75th percentiles, the whiskers are set to 5th and 95th percentile, and outliers are indicated by individual points..... 32

Figure 4: Pearson correlation (rho) over all ISMN stations included in the analysis (period 2015-2024). 33

Figure 5: Pearson correlation (rho) over selected ISMN networks. Land cover classes: see Table 3. 34

Figure 6: Box plot summarizing RMSD values for SSM calculated over all relevant ISMN stations (0-10 cm depth), per network. Data from 2015-2024 was included. The boxes indicate 25th and 75th percentiles, the whiskers are set to 5th and 95th percentile and outliers are indicated by individual points. 35

Figure 7: Box plot summarizing RMSD values per network, based on data from 2015-2023 (dark green) and 2024 (light green). Only networks with data in 2024 are included. The boxes indicate 25th and 75th percentiles, the whiskers are set to 5th and 95th percentile, and outliers are indicated by individual points..... 36

Figure 8: RMSD values over all ISMN stations included in the analysis (period 2015-2024). 37

Figure 9: RMSD values over selected ISMN networks. Land cover classes: see Table 3..... 38

Figure 10: Temporal evolution of in-situ SM (blue) and SSM1km (orange) at FMI station SOD012 at a depth=0.10m (tree cover – needle leaved). The Pearson correlation coefficient (rho) and RMSD value are given in the upper left. 41

Figure 11: Temporal evolution of in-situ SM (blue) and SSM1km (orange) at XMS-CAT station LosCoscolls at a depth=0.05m (croplands - rainfed). 41

Figure 12: Temporal evolution of in-situ SM (blue) and SSM1km (orange) at TERENO station Merzenhausen at a depth=0.05m (cropland - rainfed). 41

Figure 13: Temporal evolution of in-situ SM (blue) and SSM1km (orange) at TERENO station Selhausen at a depth = [0-0.05]m (cropland - rainfed). 42

Figure 14: Temporal evolution of in-situ SM (blue) and SSM1km (orange) at REMEDHUS station Las Eritas at depth = [0-0.05]m (cropland - irrigated). 42

Figure 15: Temporal evolution of in-situ SM (blue) and SSM1km (orange) at REMEDHUS station Carretoro at a depth = [0,0.05]m (cropland - rainfed). 42

Figure 16: Temporal evolution of in-situ SM (blue) and SSM1km (orange) at COSMOS-UK station Crichton at depth =0.10m (cropland - rainfed). 43

Figure 17: Temporal evolution of in-situ SM (blue) and SSM1km (orange) at RSMN station Iasi at depth = [0-0.05]m (cropland - rainfed). 43

Figure 18: Temporal evolution of in-situ SM (blue) and SSM1km (orange) at RSMN station Satu Mare at depth = [0-0.05]m (cropland - rainfed). 43

Figure 19: Temporal evolution of in-situ SM (blue) and SSM1km (orange) at STEMS for all four available stations (a to d) at depth = 0.10m over the same rainfed cropland field..... 44

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List of Tables

Table 1: ISMN networks used in the present SSM1km validation campaign with characteristics as given in the ISMN documentation. Note that at some stations, multiple sensors are installed at the same depth. Each of these sensors is included in the number of suitable stations. 23

Table 2: Overall summary for statistics from time series analyses of SSM1km versus in-situ measurements. Values show per network mean values of Pearson rho and RMSD. The number of included in-situ stations is given by n. (For a station to be included in this result table, it requires statistical significance defined by a Pearson p-value below 0.05) 30

Table 3: Background land cover types for Figure 5 and Figure 9, derived from the ESA WorldCover map 2020. 35

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List of Acronyms

AD	: Applicable document
API	: Application Programming Interface
ATBD	: Algorithmic Theoretical Basis Document
BfG	: German Federal Institute of Hydrology
BIEBRZA	: In-Situ Network of the Instytut Geodezji i Kartografii, Poland.
CAL/VAL	: Calibration and Validation
CEOS	: Committee on Earth Observation Satellites
CLMS	: Copernicus Land Monitoring Service
CGLOPS	: Copernicus Global Land Operations
COG	: Cloud Optimized Geotiff
DWD	: Deutscher Wetterdienst
ECMWF	: European Centre for Medium-Range Weather Forecasts
ED	: External Document
ERA5	: European Reanalysis version 5
ESA	: European Space Agency
FMI	: Finnish Meteorological Institute
FR-Aqui	: France Institute of Agricultural Research
FWC	: Framework Contract
GBOV	: Ground-Based Observations for Validation
GCOS	: Global Climate Observing System
GLDAS	: Global Land Data Assimilation System
GROW	: In-Situ Network of the GROW Observation Network
HOBE	: In-Situ Network of the Center for Hydrology – Hydrological Observatory
ICWRGC	: International Center for Water Resources and Global Change
IMA	: Institute for Agricultural and Earthmoving Machines
IPE	: Instituto Pirenaico de Ecologia
INSPIRE	: Infrastructure for spatial information in Europe
ISMN	: International Soil Moisture Network
IW	: Interferometric Wide swath mode
JRC	: Joint Research Centre of the European Union
LPV	: Land Product Validation
Metop	: Meteorological Operational Satellite
MOL-RAO	: Meteorol. Observatorium Lindenberg - Richard-Aßmann-Observatorium
NETCDF	: Network Common Data Form
Noah-LSM	: Noah Land Surface Model

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NRT	: Near Real Time
PQAD	: Product Quality Assurance Document
PUM	: Product User Manual
QAR	: Quality Assessment Report
rho	: Pearson correlation coefficient (rho)
RHEMEDUS	: In-Situ Network of the Centro Hispano Luso de Investigaciones Agrarias, Salamanca, Spain
RMSD	: Root Mean Square Difference
RSMN	: In-Situ Network of the National Meteorological Administration, Romania
SQE	: Science and Quality Evaluation
SAR	: Synthetic Aperture Radar
SM	: Soil Moisture
SQAD	: Service Quality Assurance Document
SSM	: Surface Soil Moisture
SSF	: Surface State Flag
STEMS	: In-Situ Network of the Institute of Sciences and Technologies for Sustainable Energy and Mobility
SWBM	: Soil Water Balance Model
SWI	: Soil Water Index
TERENO	: In-Situ Network of the Terrestrial Environmental Observatories
ubRMSE	: unbiased Root Mean Square Error
UNFCCC	: United Nations Framework Convention on Climate Change
VR	: Validation Report
WGS	: World Geodetic System
WMO	: World Meteorological Organization
WSMN	: Wales Soil Moisture Network

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EXECUTIVE SUMMARY

The Copernicus Land Monitoring Service (CLMS) produces a series of qualified bio-geophysical products on the status and evolution of the land surface. The products are used to monitor vegetation, crops, water cycle, energy budget and terrestrial cryosphere. The production and delivery of the parameters take place in a timely manner and are complemented by the constitution of long-term time series.

This document describes the Scientific Quality Monitoring of the 1km Surface Soil Moisture (SSM1km) products for the year 2024. In accordance with the System Quality Assurance Document (SQAD) [CGLOPS1_SQAD], the soil moisture quality monitoring within the CLMS has been transformed and further automated; the product quality analysis results are compiled as an annual update of the Quality Assessment Report (QAR).

The aim of this annual analysis is to assess the performance of the near real time (NRT) SSM1km product in the period from 2024-01-01 until 2024-12-31 relative to the performance of the SSM1km archive available from the CLMS of the previous years, starting in 2015. The validation presented in this report was performed according to the procedures defined in the Product Quality Assurance Document [CGLOPS1_PQAD].

Validation and continuous quality monitoring constitute the only means of guaranteeing the compliance of generated products with user requirements: The former concerns the new products which must pass an exhaustive scientific evaluation before to be implemented operationally. The latter concerns the operational products to check their quality keeps at the same level along the time. Both follows, as much as possible, the guidelines, protocols and metrics defined by the Land Product Validation (LPV) group of the Committee on Earth Observation Satellite (CEOS) for the validation of satellite-derived land products.

In this report, the SSM1km product, which describes in daily images the surface soil moisture in relative units (%) and is derived from Sentinel-1 Synthetic Aperture Radar (SAR) backscatter observations, is subject to quality assessment experiments. For two periods, 2015-2023 and 2024, the dataset is compared to in-situ data from suitable networks over Europe of the International Soil Moisture Network (ISMN). The analyses comprise examinations of the temporal dynamics of the SSM1km and how it agrees with the ground reference data.

Taking into account the scale gap between in-situ point measurement and the areal satellite data and the expected spatial representativeness uncertainty, the obtained results are overall as expected, with overall moderate correlation and Root Mean Square Difference (RMSD) scores, which also show a dependency on the land cover. Relatively good scores are achieved over networks in cropland and

grassland. Lower correlations, on the other hand, are mostly found in regions with forest cover or landscapes that are temporarily flooded. A seasonal bias appears at some of the validation stations, showing a partly large under- or overestimation of soil moisture during the growing season, caused by a misinterpretation of vegetation density change and vitality dynamics in the radar backscatter modelling. Generally, it was found that the SSM1km has a good skill to reproduce peaks and lows of the ground soil moisture.

Additionally, the users may refer to the dedicated peer-reviewed publication of Bauer-Marschallinger et al. (2018), where the authors provide an in-depth description of the retrieval methodology, and offer also spatio-temporal evaluation experiments at the 1km scale, and demonstrate the dataset's skill to capture rainfall patterns. A short summary of this study is given in Section 6.1.

The comparison of data from the year 2024 versus data from 2015-2023 showed overall stability of the product quality. Except for stations with extraordinary climate events (drought, cold winters), we see a slight quality improvement over every analysed network, also confirming what can be expected from the stable quality of the Sentinel-1 input data in the recent years, despite the end of Sentinel-1B observations resulting from the satellite failure in December 2021. The resulting reduced observational frequency during the 2022-2025 period might lead to missed short-term events, as e.g. from local thunderstorms.

Sentinel-1A is stably observing signals related to soil moisture, and return to a regular coverage from a full two-satellite-constellation is expected for 2025 as Sentinel-1C was launched successfully at the end of 2024. By begin of May 2025, the Sentinel-1C observation scenario is already close to the nominal operations, and the S-1C commissioning activities to achieve again nominal S-1 mission coverage are ongoing and expected to finish mid-2025.

1 BACKGROUND OF THE DOCUMENT

1.1 SCOPE AND OBJECTIVES

The document presents the results of the annual scientific quality monitoring of Surface Soil Moisture 1km Version 1 product (named hereafter SSM1km) derived from Sentinel-1 radar backscatter images, acquired in Interferometric Wade Swath (IW) mode and VV-polarization (see ED1 for details). This product provides daily soil moisture images over Europe from January 2015 ongoing. The quality evaluation is performed on SSM1km data from 2024-01-01 to 2024-12-31 with a reference period from 2015-01-01 to 2023-12-31.

The objective is to check that the operational products keep the same level of quality in the period under study than the fully validated products.

1.2 CONTENT OF THE DOCUMENT

This document is structured as follows:

- Chapter 2 recalls the requirements, and the expected performance
- Chapter 3 reviews the results of previous validation reports
- Chapter 4 describes the methodology for quality assessment, the metrics and the criteria of evaluation
- Chapter 5 presents the results of the analysis
- Chapter 6 summarizes the main conclusions of the study
- Chapter 7 makes recommendations based upon the results

1.3 RELATED DOCUMENTS

1.3.1 Applicable documents

AD1: Part 2: Technical specifications of Framework Service Contract – Operation of the bio-geophysical variables systematic monitoring of the Global Land Component of the Copernicus Land Service 'CGLOPS' JRC/2023/OP/0273, 19th April 2023.

Available at https://etendering.ted.europa.eu/cft/cft-display.html?cftId=13795

1.3.2 Input

Document ID	Descriptor
CGLOPS1_PQAD	Product Quality Assurance Document for global Vegetation and Energy products of the Copernicus Land Monitoring Service.
CGLOPS1_SQAD	System Quality Assurance Document for global Vegetation and Energy products of the Copernicus Land Monitoring Service.
CGLOPS1_ATBD_SSM1km-V1	Algorithm Theoretical Basis Document of the Surface Soil Moisture Version 1 product.
CGLOPS1_VR_SSM1km-V1	Validation Report for the Surface Soil Moisture 1km Version 1 product.
CGLOPS1_VR2024_SWI1km	Validation Report of the Soil Water Index 1km for 2024.

Available at: https://land.copernicus.eu/en/technical-library

1.3.3 Output

Document ID	Descriptor
CGLOPS1_PUM_SSM1km-V1	Product User Manual summarizing all information about the Surface Soil Moisture Version 1 product.

Available at: https://land.copernicus.eu/en/technical-library

1.3.4 External Documents

Document ID	Descriptor
ED1	Sentinel-1 User Guide

https://sentiwiki.copernicus.eu/web/document-library#Library-S1-Documents

GCOS-245	The 2022 GCOS ECVs Requirements

https://library.wmo.int/records/item/58111-the-2022-gcos-ecvs-requirements-gcos-245?offset=1

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2 REQUIREMENTS

According to the applicable documents [AD1], the requirements relevant for SSM1km are described below.

2.1 SPECIFIC TECHNICAL DETAILS AND REQUIREMENTS

Surface Soil Moisture (SSM) is the relative water content of the top few centimetres soil, describing how wet or dry the soil is in its topmost layer, expressed in percent saturation.	
Geometric Properties:	
spatial resolution	1km
geolocation precision	Better than 0.5 pixels
Coordinate position	Centre of the pixel
Geodetical datum	WGS84
Geographic projection	regular latitude/longitude grid
Geographic Coverage:	EU
Time span	Daily
Timeliness	12 hrs
Uncertainty (2-sigma)	Threshold: 20%; Goal: 10%
Stability (per decade, i.e. 10 years)	Threshold: 5% Goal: 3%

2.2 FURTHER REQUIREMENTS

2.2.1 Output product composition

Products may contain various information layers and ancillary information – the base reference for product packages are the operational products as on 01.03.2023.

2.2.2 Data Structure

Data coding[1] shall be compatible with the Global Land products as on 01.03.2023 and/or follow the INSPIRE specifications, where applicable.
Ancillary information shall be as currently used and include at least the following:

- The number of measurements per pixel used to generate any synthesis product
- The per-pixel date of the individual measurement or the start-end dates of the period covered
- Quality indicators, with explicit per-pixel identification of the cause of anomalous parameter result.

The product naming and filename conventions that are used in the Copernicus Global Land component production as on 01.03.2023 shall be followed. This may be adapted for complete product collections upon agreement with the contracting authority during the Framework.

2.2.3 Data format

To ensure interoperability with the current Global Land component (operational product data formats and archive data formats) and other Copernicus services, all datasets will be available in NETCDF. Additional format such as Cloud Optimized Geotiff (COG) or ZAR format can be proposed for production or could be requested by the Contracting Authority during the Framework Contract.

2.2.4 Uncertainties and Validation

Uncertainties indicated in the product specifications above follow the threshold and goal proposed by GCOS#245.

Uncertainties estimates should account for the error propagation uncertainty coming from input data though the retrieval algorithms as during the contract period, ESA plans to imbed uncertainties in the Sentinel-3 ground segment products and, for Sentinel-2, there is an offline tool to determine Evel 1 uncertainties; these can be used for propagation in the production chain.

Validation of the products shall conform to at least the CEOS LPV standards. Wherever appropriate the bio-geophysical variables shall be validated and compared to CEOS CAL/VAL data sets and/or Ground-Based Observations for Validation (GBOV) of Copernicus Global Land component when biophysical parameters are available.

[1] Data coding is the provision for the number of bits, range of values, usage of reserved values, content of status map, conversion to physical values formulas.

2.2.5 Input data

Copernicus sentinel data are available from https://dataspace.copernicus.eu

Bio-geophysical variable products should be based on common base reflectance data:

- Sentinel 3: Derived Top-of-Canopy reflectance may be brokered or produced as it is the case on 01.03.2023 under the CGLOPS contracts;
- Sentinel 2: the Global Land Sentinel-2 Global Mosaic (S2GM)[2] component provides temporal mosaic of surface spectral bands that can be brokered and/or directly used.

Up to and including 2019, products archives of the Global Land component have been based on SPOT VGT, Proba-V, ENVISAT, MODIS, TOPEX/Poseidon, Jason-1, Jason-2, Jason-3, datasets, which are available through the https://dataspace.copernicus.eu.

LST and SWI can be based on geostationary and other satellite data.

Ancillary satellite data that is purchased through Copernicus and put at disposal of the Services is available through the Data Warehouse and will become available on the Copernicus Dataspace Ecosystem.

Ancillary data sets, other than satellite imagery described above, that might be required shall be the responsibility of the contractor.

2.2.6 Product delivery

Products shall be delivered to the Copernicus Land Component dissemination. The Copernicus Data Space Ecosystem infrastructure will be used to provide access to the final map products.

[2] https://land.copernicus.eu/en/products/global-image-mosaic

3 REVIEW OF INITIAL SSM1KM VALIDATIONS

In 2017, an initial Validation Report [CGLOPS1_VR_SSM1km-V1] was produced to assess the quality of the Copernicus Land Monitoring Service's SSM1km product, which describes in daily images the surface soil moisture in relative units (%) and is derived from Sentinel-1 Synthetic Aperture Radar (SAR) backscatter observations. For the year 2015, the dataset was compared to data from six in-situ networks over Europe accessed from the International Soil Moisture Network (ISMN). The quality of the SSM1km was assessed in the temporal and in the spatial domain, meaning that the quality of the temporal signal and the spatial signal was analysed individually. In the remainder of this section, the conclusions from the Validation Report (VR) are revisited.

The temporal analyses presented poor to medium performances of the product when compared against in-situ data, albeit the quality of the in-situ data itself was not questioned. While there was acceptable agreement with stations in Austria, Germany, Romania and Finland, little agreement was found with data from Spain and France. Due to the temporal frequency of the SSM1km, which is eminently inferior to the one of the in-situ observations, the temporal signal was inevitably of lower quality. Many events, leading to peaks in the in-situ data, were missed by the satellite product. This was considered as the most impacting drawback of the SSM1km. However, the overall level and long-term development of soil moisture were mostly captured. Putting the study in relation to similar validation campaigns, the obtained correlation results undercut only little the results from other remotely sensed soil moisture data.

The spatial analyses show in contrast a much higher agreement, with almost coincident soil moisture patterns in the data from in-situ and SSM1km in the Spain-, Finland- and France- validation sites. Daily soil moisture maps agreed very well, meaning that in the network area, the relative soil moisture differences at a time were well captured by the satellite data. This is not unexpected, as this accords with high quality of the Sentinel-1 image data due to its spatial detail. The product showed higher potential to describe the variation of soil moisture in space than other remotely sensed datasets.

The SSM Water Mask and the SSM Sensitivity Mask showed good results when compared against land cover data. The Sensitivity Mask performed well and managed to identify city areas, but left some room for improvement. The SSM Water Mask does not miss any water pixel, practically, but only overestimates the water area slightly, which is rather a safeguard to the SSM product.

Overall, the SSM1km met the expectations as it achieved to reflect the spatial dynamics of soil moisture at the kilometric scale. On that matter, the product shows more potential than comparable products. However, the performance over time against time series data from in-situ point measurements was rather weak and left room for improvement.

In December 2018, a peer-reviewed publication (Bauer-Marschallinger et al., 2018) explained first in detail the SSM1km retrieval algorithm and then investigated its performance over Italy and its neighbours. Over that area, which comprises challenging terrains and heterogeneous land cover, a consistent set of model parameters and product masks was generated, unperturbed by coverage discontinuities. An evaluation of therefrom generated SSM1km data, involving in-situ measurements and a 1km Soil Water Balance Model (SWBM) over Umbria, yielded high agreement over plains and agricultural areas, with low agreement over forests and strong topography. Also here, positive biases during the growing season were detected. However, excellent capability to capture small-scale soil moisture changes as from rainfall or irrigation was evident from this evaluation.

In the assessment for the year 2021, the impact from the failure of Sentinel-1B in December 2021 (with Sentinel-1A remaining operational) was analysed. It was found that the temporal resolution of the SSM1km product is about halved as long as the successor Sentinel-1C will be operational. The SSM1km model parametrisation and the available SM observations from Sentinel-1A appeared to be unaffected in terms of quality. Since the Sentinel-1C satellite was launched in December 2024, the effects of restored resolution can only be seen in 2025.

In the year 2022, the performance of the near real time (NRT) SSM1km product was assessed against the previous multiyear data, and the report showed overall a stable and partially improving data quality. Similar patterns were found in 2023.

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4 SCIENTIFIC QUALITY MONITORING METHOD

In this study, the SSM1km product of the CLMS is evaluated on its signal quality and stability. The SSM1km retrieval is explained in detail in [CGLOPS1_ATBD_SSM1km-V1], and usage- and format-specifications are discussed in [CGLOPS1_PUM_SSM1km-V1].

4.1 IN-SITU REFERENCE PRODUCTS

For in-situ validation purposes, the project consortium can access the comprehensive database of the ISMN (International Soil Moisture Network; Dorigo et al., 2011, Dorigo et al, 2021). It is an international cooperation to establish and maintain a global in-situ Soil Moisture (SM) database, and is recommended by CEOS' Soil Moisture Product Validation Good Practices Protocol (Montzka et al., 2020).

This database is an essential means of the geoscientific community for validating and improving global satellite observations and land surface models. As of March 2025, filtered for the period 2015 to 2024, the ISMN contains data of 85 networks and more than 4000 stations located all around the globe, with highest density over Europe and North America, and most datasets developed during the last two decades. The database is rapidly expanding, which means that both the number of stations and the period covered by the existing stations are still growing. Hence, it has been become an increasingly important resource for validating and improving satellite-derived soil moisture products and studying climate related trends. As the ISMN is maintained by the scientific community itself, potential networks are invited to enrich the collection by sharing its in-situ soil moisture data.

The ISMN employs an automated quality control system for its SM measurements, which flags values exceeding a certain threshold and checking the validity of SM variations in relation to changes in soil temperature and precipitation (Dorigo et al., 2013). A second flagging procedure analyses the shape of the SM time series to detect outliers, positive and negative breaks, saturation of the signal, and unresponsive sensors. In this analysis, only values with the flag "G" (good) are kept. Dorigo et al. (2021) reported on the recent developments within the ISMN and provide a detailed documentation on the network update activities and data quality curation. This includes an ongoing activity on quality curation of historic time series that aims for most consistent and reliable in situ data. Consequently, the available time series data at ISMN may be altered over time in terms of flagging and value density. To ensure a long-term funding, the ISMN operations were transferred by the end of 2022 to the International Center for Water Resources and Global Change (ICWRGC) and the German Federal Institute of Hydrology (BfG), and both new hosting intuitions are dedicated to continue ongoing activities on network integration and quality check.

4.1.1 Used ISMN Networks

For the validation of the SSM1km dataset, the ISMN was browsed for networks that fulfil the following requirements:

- Location in the SSM1km coverage area, i.e. the Europe production window
  - bottom left: -12°E, 35°N, top right: 45°E, 72°N
- Valid observations during the validation period Jan 2015 – Dec 2024
  - First, a suitable station needs at least one year of in-situ data available to be considered
  - After temporal matching with SSM1km and masking of cold temperatures, at least 30 observations per station must be available.
  - As further criteria on the ISMN input data density, results in Section 5 are only plotted with data from locations, where the correlation has a high significance with a p-value lower than 0.05.
- SM measurement of near-surface layer (shallower than 10 cm)

The ISMN networks are constantly updated, either via an Application Programming Interface (API) automatically refreshing the data in near-real-time, or via manual updates. Some networks have update delays and were not updated for the entire 2024 at the time of the validation. The ISMN data was accessed the 15th of April 2025. Comparing to last year's report, the addition of the German Weather Service (DWD) network with Sentek Drill and Drop sensors shall be highlighted. This network provides in-situ surface soil moisture covering Germany with 20 stations (only 11 stations suitable for this year's analysis). Additionally, the network WATERLINE in Greece with 3 stations (only 2 used for analysis) with Sentek Drill and Drop sensors used for surface soil moisture was also used. While the two networks (COSMOS and COSMOS-UK) did not get updated for 2024, their use of cosmic ray neutron sensors provides higher accuracy data for comparison with SSM1km product.

In terms of covered period, the validation of the SSM1km is twofold: 1) For the long-term validation, every sensor with data between 2015 and (inclusive) 2024 are used; 2) when comparing the product quality for 2024, the networks and sensors are filtered to only use the ones with SSM available for the period of 2024, leading to a smaller set of analysed networks.

A detailed overview of the locations of all used ISMN stations fulfilling the requirements delineated above is shown in Figure 1 and represent the area covered by SSM1km well. For some networks, a larger number of stations than indicated in the "suitable stations" column is available via ISMN. However, some of these stations only measured data for a few weeks, which hampers the robust calculation of validation metrics. During the temporal collocation of SSM1km and in-situ

measurements, the number of measurements is reduced even further due to the relatively coarse temporal resolution of SSM1km, especially before the launch of Sentinel-1B in 2016, and after its failure in end of 2021. In order to ensure a reliable derivation of evaluation results, only stations with at least 30 collocated SSM1km and in-situ observations are kept. This follows the suggestion drawn from the Central Limit Theorem of probability theory for parametric tests like Pearson correlation.

For more information on the ISMN network details and contacts, please visit: https://ismn.earth/en/networks/.

Table 1: ISMN networks used in the present SSM1km validation campaign with characteristics as given in the ISMN documentation. Note that at some stations, multiple sensors are installed at the same depth. Each of these sensors is included in the number of suitable stations.

Network Name	Data Availability	Climate (Koppen-Geiger)[3]	Altitude [m]	Number of suitable stations (0-10 cm depth)
BIEBRZA_S-1 Poland	2015-04-23 to 2018-12-01	Dfb	110-112	13
COSMOS Austria/Switzerland	2010-12-13 to 2017-04-05	Cfb, Dfb	270-750	2
COSMOS-UK Great-Britain	2013-03-10 to 2023-01-01	Cfb	3-365	131
DWD Germany	2024-01-01 to 2024-12-31	Cfb, Dfb	3-877	11
FMI northern Finland	2007-01-25 to 2024-12-31	Dfc	150-441	25
FR_Aqui France	2012-01-01 to 2023-12-31	Cfb	25-54	7
GROW Europe	2017-02-08 to 2019-10-08	Cfb, Dfb	European network	122
HOAL Austria	2013-07-11 to 2021-12-31	Dfb	270-321	63
HOBE Denmark	2009-09-08 to 2019-03-13	Dfb	26-104	112

[3] **Koppen-Geiger climate information :**
1st letter : B (dry), C (temperate), D (Continental), E (Polar)
2nd letter : w (Dry winter), f (No dry season), s (Dry summer),
3rd letter: a (Hot summer), b (Warm summer), c (Cold summer), k (Cold for dry climate only), T (Tundra)

Network Name	Data Availability	Climate (Koppen-Geiger)[3]	Altitude [m]	Number of suitable stations (0-10 cm depth)
HYDROL-NET-PERUGIA Italy	2010-01-01 to 2016-12-31	Csa	285	2
IMA_CAN1 Italy	2011-08-23 to 2015-12-03	Cfa	272-278	6
IPE northern Spain	2008-04-03 to 2020-03-25	Cfb	750	2
LABFLUX Italy	2020-04-23 to 2023-06-19	Dfb	640	2
MOL-RAO eastern Germany	2003-01-01 to 2020-06-30	Dfb	49-73	2
REMEDHUS Spain	2005-03-15 to 2024-12-31	Bsk	890-720	20
RSMN Romania	2014-04-09 to 2024-12-31	Cfa, Dfa, Dfb, Dfc	19-318	20
SMOSMANIA southern France	2007-01-01 to 2024-01-01	Cfa, Cfb, Cfc, Csa, Dfa	30-1240	46
STEMS northern Italy	2015-12-04 to 2024-12-31	Cfa	275	4
TERENO western Germany	2009-12-31 to 2024-12-31	Cfb	70-612	14
TWENTE Netherlands	2008-19-11 to 2020-12-31	Cfb, Dfb	0-49	71
UMBRIA central Italy	2002-10-09 to 2017-12-31	Csa, Cfa	215-550	8
UMSUOL Italy	2009-06-12 to 2017-05-15	Cfa	11	1
WATERLINE Greece	2021-10-06 to 2024-09-09	Csb, Dsb	34-711	2
WSMN Great Britain	2011-09-02 to 2016-02-29	Cfb	28-477	4
XMS-CAT eastern Spain	2016-08-01 to 2024-12-31	Dfb, Dfc, Csb	38-1236	16

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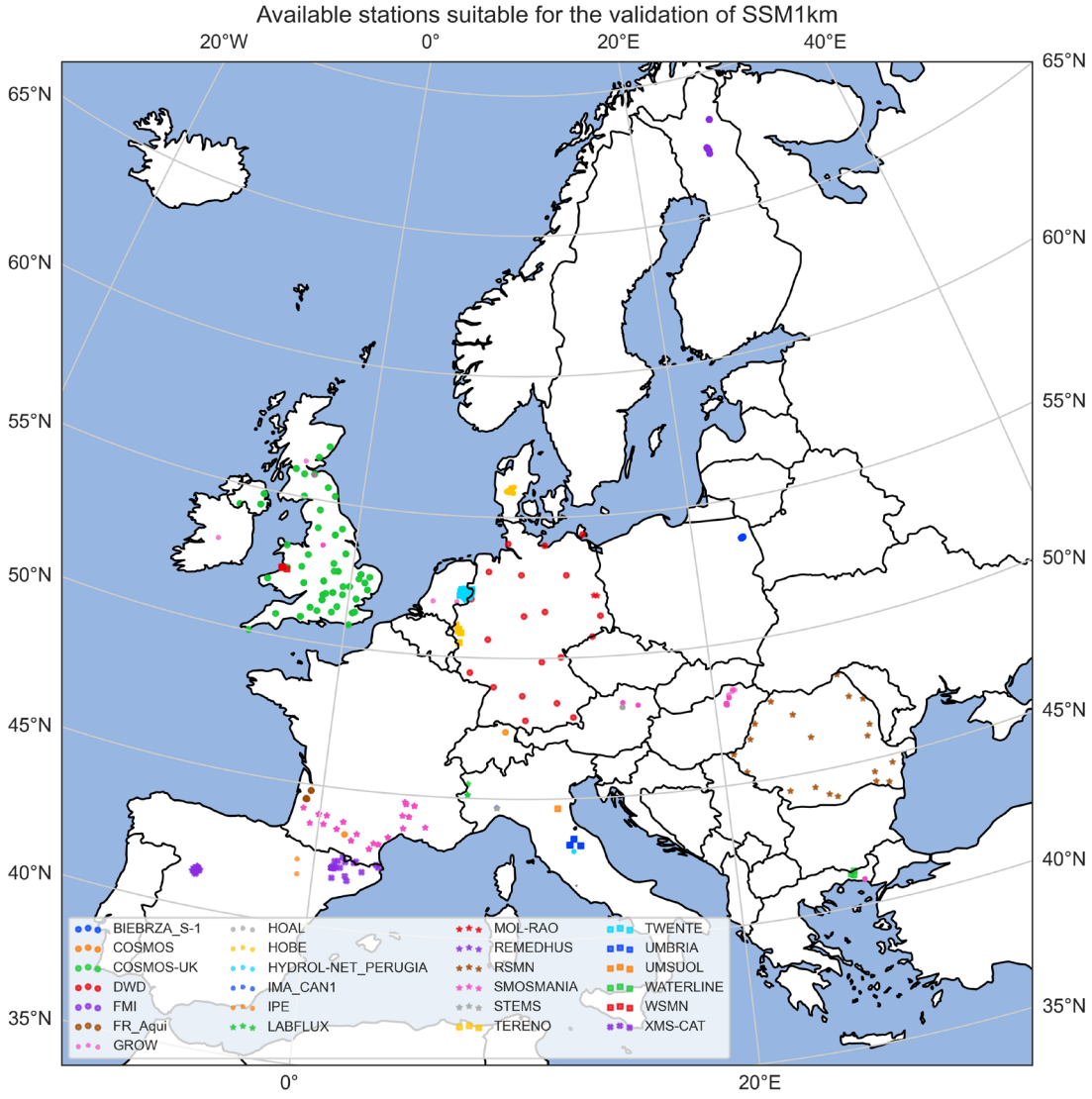


Figure 1: Spatial distribution of ISMN stations used for the present SSM1km validation.

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4.1.2 Spatial Representativeness

For the validation of high-resolution soil moisture products as the SSM1km, actual ground measurements from in-situ stations are of major importance. Such measurements are able to represent soil moisture variations at point- or field-scale, depending on the location's area homogeneity and the sensor design. When comparing remotely sensed and in situ data, one must consider spatial representativeness. This means how well an in-situ observation, which is essentially a point measurement, represents the data gathered by the much larger satellite footprint. This is in general different for each station and depends on topography, soil types and microclimates. De Lannoy et al. (2007) discussed relationship between point measurements and the average behaviour of field-scale soil moisture and concluded that the choice of individual sites has considerable impact on validation campaigns, suggesting field-level analysis of the local spatial SM signal structure and its drivers, followed by upscaling procedures.

For this validation, all datasets are kept with their native product samplings as an in-depth analysis of in-situ sites is out of scope in this endeavour. Consequently, results from this validation always need to be interpreted in reflection of the large difference of scales. In the case of the present validation study of 1 km data, this so-called representativeness error is assumed not to be negligible and thus validation results need to be reflected accordingly. Crow et al. (2012) stated that during validation activities based on comparisons between point-like ground observations and extensive satellite retrievals, this sampling error can be misattributed to retrieval uncertainty and spuriously degrade the perceived accuracy of satellite soil moisture products. They found that standard deviations of SM in sample sets spanning 0.8 km² and 1.6 km² are of the same or larger magnitude as the CEOS' accuracy criterion for SM of 0.04 m³/m³. Gruber et al. (2013) carried out an in-depth analysis of coarse-scale representativeness of ISMN networks and they found absolute error estimates ranging from 0.02 to 0.06 m³/m³. However, one can expect general lower errors when analysing SM anomalies and relative dynamics.

Alternatively, a validation against modelled SM data of the same scale as the satellite product could be done. However, for a continentally or globally comprehensive validation of remotely sensed SM at kilometric scale, there is a lack of comparable extensive datasets with similar spatial resolution and extent. Fortunately, regional model output at kilometric scale describing SM, even with limited extent, can provide an accurate assessment of the temporal and spatial SSM signal quality. The reader may here refer to the study of Bauer-Marschallinger et al. (2018) where the authors examined the SSM1km dataset against an extensive 1 km soil moisture model dataset covering the Umbria region in Italy.

4.2 VALIDATION PROCEDURE

The aim of this validation report is to assess the performance of the SSM1km product in the period from 2015-01-01 to 2024-12-31 with respect to ground measurements at the stations collected by the ISMN. For each location, SSM1km time series are extracted and statistically compared to time series from the in-situ stations.

4.2.1 Dataset Preparations

4.2.1.1 Temporal Resampling

The SSM1km datasets are available at the times of the respective Sentinel-1 overpass, leading to observations approximately every 1.5-4 days over Europe with two satellites active (until December 2021). From end of 2021 onward, one satellite is active. The reference in-situ datasets offer generally a much higher temporal sampling and are thus down-sampled for this validation. Therefore, the in-situ SM values are first averaged to half-daily timestamps and then, the timestamp that is closest in time to the SSM1km timestamp is selected. The daily SSM1km images are converted to time series through co-registered stacking along the time dimension.

4.2.1.2 Masking for Invalidating Conditions

SSM1km and in-situ time series are temporally masked for invalid observations, where soil moisture measurements are prone to be erroneous. Typical conditions are frozen soil and snow. In this analysis, we use the quality flagging of the ISMN in-situ data, the flags for snow coverage, frozen soil, saturated signal plateaus, and excessive precipitation (see Section 4.1).

The ISMN team announced on January 16th, 2025 that the algorithm for data quality flagging was updated and improved. These updates mainly affected three of the quality flags (D03, D05, and D10) on the following NRT networks (FMI, RSMN, TERENO) and the regular networks (COSMOS, FR_Aqui, HOAL, LABFLUX, REMEDHUS, STEMS, and WATERLINE) used in our analysis. It should be noted that a changing dataset sample size due to ISMN flag updates – and related shifts in mean and standard deviation – can cause different appearance of the scaled SSM1km data in the different annual QAR reports (or even alter the decision if the location falls below the threshold of 30 matching observations). It should be noted that the changes in flagging of ISMN data might have affected the values in the data for years previous to 2024. However, no discernible effect on the data for this year validation could be seen.

4.2.1.3 Temporal Matching

For the time series comparison, the individual datasets are checked at each validation location for no-data-values. Then both time series are time-stamp-matched, reducing them to timestamps where both time series have valid data. Two timestamps are matched if they co-locate within a 12 hours temporal interval. This allows for the calculation of Pearson correlation coefficients and Root Mean Square Difference (RMSD) values as measures of agreement.

4.2.1.4 Data Scaling

Since the SSM1km data is provided in percentage of relative saturation (i.e. %), while the reference products are provided in m^3/m^3 (ranging commonly from 0.0 – 0.5 m^3/m^3), data scaling is necessary prior to the evaluation of difference metrics such as RMSD. Following leading experts and current literature, scaling into the in-situ data space is done (Brocca et al., 2011, Montzka et al, 2020).

In order to report here the targeted metrics in m^3/m^3, "mean-standard-deviation scaling" (also known as z-normalisation) is used to scale the SSM data to the reference products (Lima and Souza, 2023). With this method, the SSM data is scaled to have the same mean value and standard deviation as the reference dataset. This approach was chosen because it is widely adopted by the community and because the extent of the scaled values is less affected by the presence of outliers, compared to for instance a "minimum-maximum scaling". Moreover, through the mean-standard-deviation scaling, no statistical bias remains between CGLS and reference time series, and the obtained RMSD values are equivalent to the unbiased Root Mean Square Error (ubRMSE), a metric regularly used within good practices established by the CEOS LPV (Montzka et al, 2020).

4.2.2 Assessment of the Temporal Signal Quality

For the assessment of temporal signal quality, the SSM1km data is compared to data from the selected ISMN networks. For the comparison, basic statistical measures are calculated: the Pearson correlation coefficient (rho) and the Root Mean Square Difference (RMSD).

The obtained validation metrics are presented and discussed in Section 5. First, an overview of the overall SSM1km dataset quality is given per network, along with an assessment of quality differences between the SSM1km dataset in the year 2024 and in the reference period 2015-2023. This is complemented by an in-depth analysis over selected individual ISMN stations, discussing the temporal dynamics of SSM1km and in-situ SM time series and their agreement.

5 RESULTS

This section first presents the mean values of the performance indicators Pearson rho and RMSD per network, then shows summary box plots and network overviews per soil depth layer, and features in the end time series plots to illustrate selected examples of good and poor agreement between SSM1km and in-situ.

5.1 OVERALL ANALYSIS SUMMARY

The results of the statistical comparison of SSM1km against ISMN in-situ stations are overall satisfying, with a similar performance at ISMN networks all over Europe (Table 2). The weighted means (with the number of stations used as weight) show correlation similar compared to last year.

An overall average **rho of 0.39** and overall average **RMSD of 0.08** m^3/m^3 are obtained when weighting the metrics with the respective number of stations. Please note that the results have a large variance among the networks, as over certain networks the dataset quality is better than the average. For example, the maximum average rho of over 0.50 is obtained over the STEMS, WSMN and COSMOS-UK network. RMSD values of 0.05 or below are observed over STEMS, RSMN, and IMA-CAN1. Moreover, the performance also varies among the stations of each network due to differences between the specific sites, e.g., related to land cover and soil composition. This year, two networks show even negative averaged correlation, namely IPE and HYDROL-NET-PERUGIA. IPE consists of two stations and the negative averaged correlation is caused by the station Aguero, located in a dense and hilly forest in the province of Aragon, showing a correlation of -0.34. The HYDROL-NET-PERUGIA network consists of two stations within the same Sentinel-1 pixel and provides only two years of overlapping data with the S1 SSM product. The stations are located in a forest patch surrounded by urban areas, which might explain the obtained low correlation values.

The obtained performance metrics are consistently lower than for the SWI1km product (see [CGLOPS1_QAR2024_SSM1km]). This can be explained mainly by the fact that SWI1km uses both ASCAT and Sentinel-1 data as input. Thus, the temporal behaviour, which is to a large extent based on ASCAT data, is improved compared to the SSM1km product, which reflects in the performance metrics. Large differences between SSM1km and SWI1km's correlation and RMSD are also observed by very low number of observation and incomplete dataset (e.g. HYDROL-NET-PERUGIA with only 18 months of data in the early phase of Sentinel-1 (until 2016)). The results of the statistical comparison are presented in more details in the following sections.

Table 2: Overall summary for statistics from time series analyses of SSM1km versus in-situ measurements. Values show per network mean values of Pearson rho and RMSD. The number of included in-situ stations is given by n. (For a station to be included in this result table, it requires statistical significance defined by a Pearson p-value below 0.05)

Network	Validation metrics		
	n	Ø rho	Ø RMSD [m³/m³]
BIEBRZA_S-1	13	0.23	0.10
COSMOS	2	0.13	0.06
COSMOS-UK	131	0.50	0.09
DWD	11	0.43	0.09
FMI	25	0.04	0.06
FR_Aqui	7	0.37	0.07
GROW	122	0.46	0.08
HOAL	63	0.41	0.07
HOBE	112	0.30	0.07
HYDROL-NET_PERUGIA	2	-0.23	0.09
IMA_CAN1	6	0.42	0.04
IPE	2	-0.06	0.09
LABFLUX	2	0.23	0.07
MOL-RAO	2	0.14	0.06
REMEDHUS	20	0.44	0.06
RSMN	20	0.40	0.05
SMOSMANIA	46	0.29	0.08
STEMS	4	0.51	0.05
TERENO	14	0.39	0.09
TWENTE	71	0.47	0.09
UMBRIA	8	0.30	0.07
UMSUOL	1	0.34	0.09
WATERLINE	2	0.22	0.07
WSMN	4	0.51	0.06
XMS-CAT	16	0.33	0.09
Total	***706***		
Mean (networks weighted by stations number)		**0.39**	**0.08**

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5.1.1 Summary over Networks: Pearson Correlation Analyses

In the following, box plots show the distributions of the Pearson rho values per network, along with the number of stations with in-situ data for the evaluation period 2015-2024 (Figure 2). The temporal extent of available data for the different networks is described in Table 1. Only the stations with a significant result and a Pearson p-value below 0.05 are used for the box plots. Please note, in these figures, boxes are replaced by horizontal lines in case of a single station, and that the values in Table 2 are mean values, whereas the distributions in the box plots mark median values.

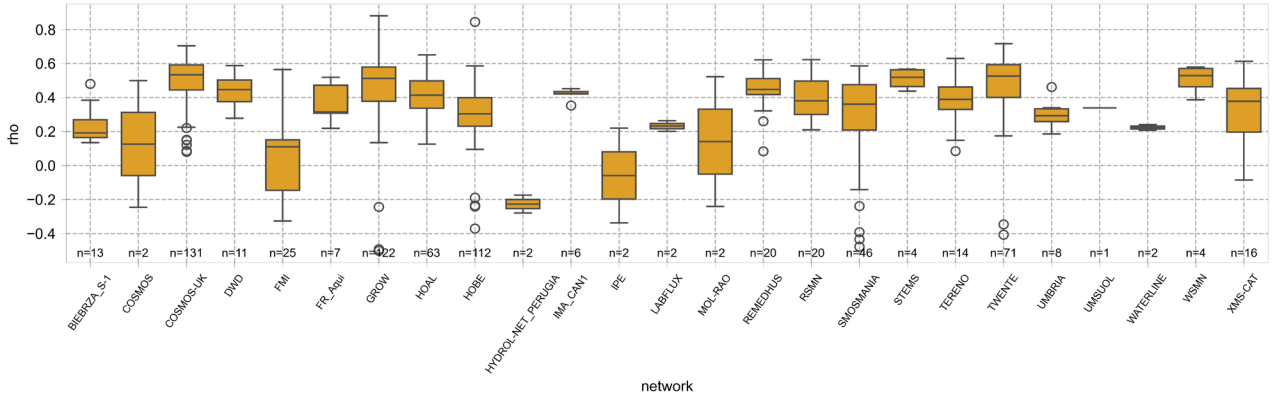


Figure 2: Box plot summarizing Pearson rho values for SSM calculated over all relevant ISMN stations (0-10 cm depth), per network. Data from 2015-2024 was included. The boxes indicate 25th and 75th percentiles, the whiskers are set to 5th and 95th percentile and outliers are indicated by individual points

Many networks show moderate to good scores with median rho values ranging from around 0.25 to 0.55. At individual stations, rho values range up to about 0.80, and some down to -0.50. When only a small number of stations are available in a network, and the obtained metrics differ largely among the stations, the percentile boxes as well as the 5th and 95th whiskers have a large spread. (e.g. MOL-RAO, FMI, IPE, COSMOS). Details on the performance at selected individual stations are provided in Section 5.2.

Next, the Pearson rho values obtained when using only data from the year 2024 are compared to the obtained rho values for the period 2015-2023 (Figure 3). Here, only networks that measured soil moisture consistently in 2024 (with minimum 30 observations and a p-value below 0.05 for 2024) have been included in the analysis, narrowing the set of analysed stations.

Over the networks FMI, REMEDHUS, RSMN, STEMS, and XMS-CAT the median correlations in 2024 are either similar or much higher than in the reference period (Figure 3). Over TERENO and WATERLINE the correlation is lower compared to the previous years. In the case of RSMN, some stations have not been fully updated (e.g. Rosioriide Vede), or show a discontinuous/erroneous SSM

signal (e.g. Chisineu Cris). Multiple in-situ stations over TERENO shows high plateaus of SSM with regards to previous years (Gevenich, station-C). These plateaus are not fully captured by the Sentinel-1-based SSM signal, showing a slight correlation reduction (some stations, such as Selhausen, station-A, even show discontinuous data). Concerning XMS-CAT located in Catalonia/Spain, comparisons at some stations results are potentially negatively impacted by elevation and rough terrains. As an example, the station Pessonada at 900m elevation, contains cliffs and a 500 m elevation gain within the Sentinel-1 pixel. As result, correlation for this station is -0.14, and for 2024, lowered to -0.08. Overall, the results meet expectations, as the overall quality of the input from Sentinel-1 has increased since mission start, in terms of radiometric accuracy and temporal coverage.

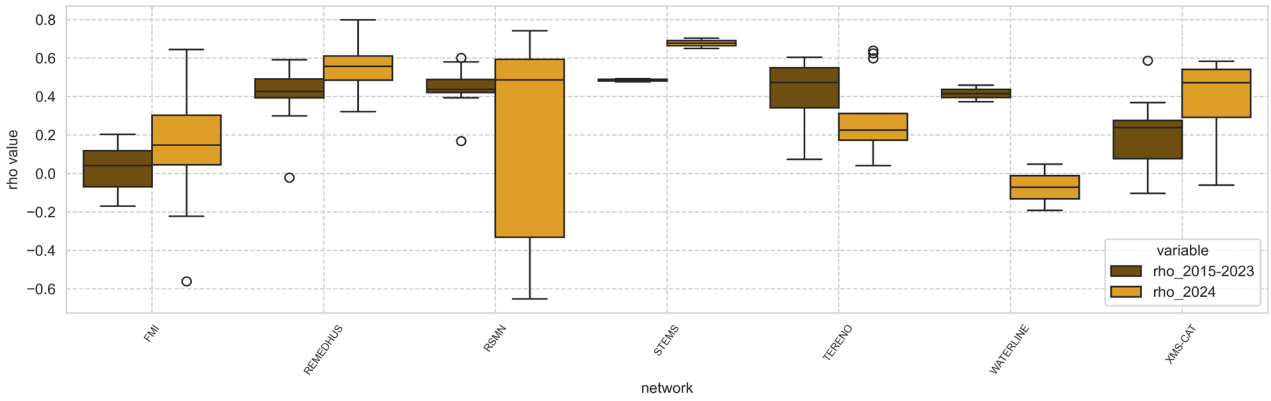


Figure 3: Box plot summarizing Pearson rho values per network, based on data from 2015-2023 (brown) and 2024 (orange). Only networks with data in 2024 are included. The boxes indicate 25th and 75th percentiles, the whiskers are set to 5th and 95th percentile, and outliers are indicated by individual points.

The failure of Sentinel-1B reduced the number of available observations for the period of 2022 - 2024, thus reducing the number of points available for the statistical analysis. Apart from the aforementioned TERENO and WATERLINE networks, the product's correlation to in-situ data remains as high/higher in 2024 compared to previous years. Similar to its performance in 2023, FMI shows a higher median Pearson correlation compared to previous years but remains slightly lower compared to the other networks. Previous analyses suggest that this is due to the challenging land cover type and cold climate conditions with a highly dynamic freeze/thaw-processes. The WATERLINE network's correlation in 2024 suffers from lack of data for three months out of the year, as only data until September is available. Additionally, in-situ soil moisture dynamics display high plateaus that are not captured by the satellite SSM, likely due to the rough topography of the sites.

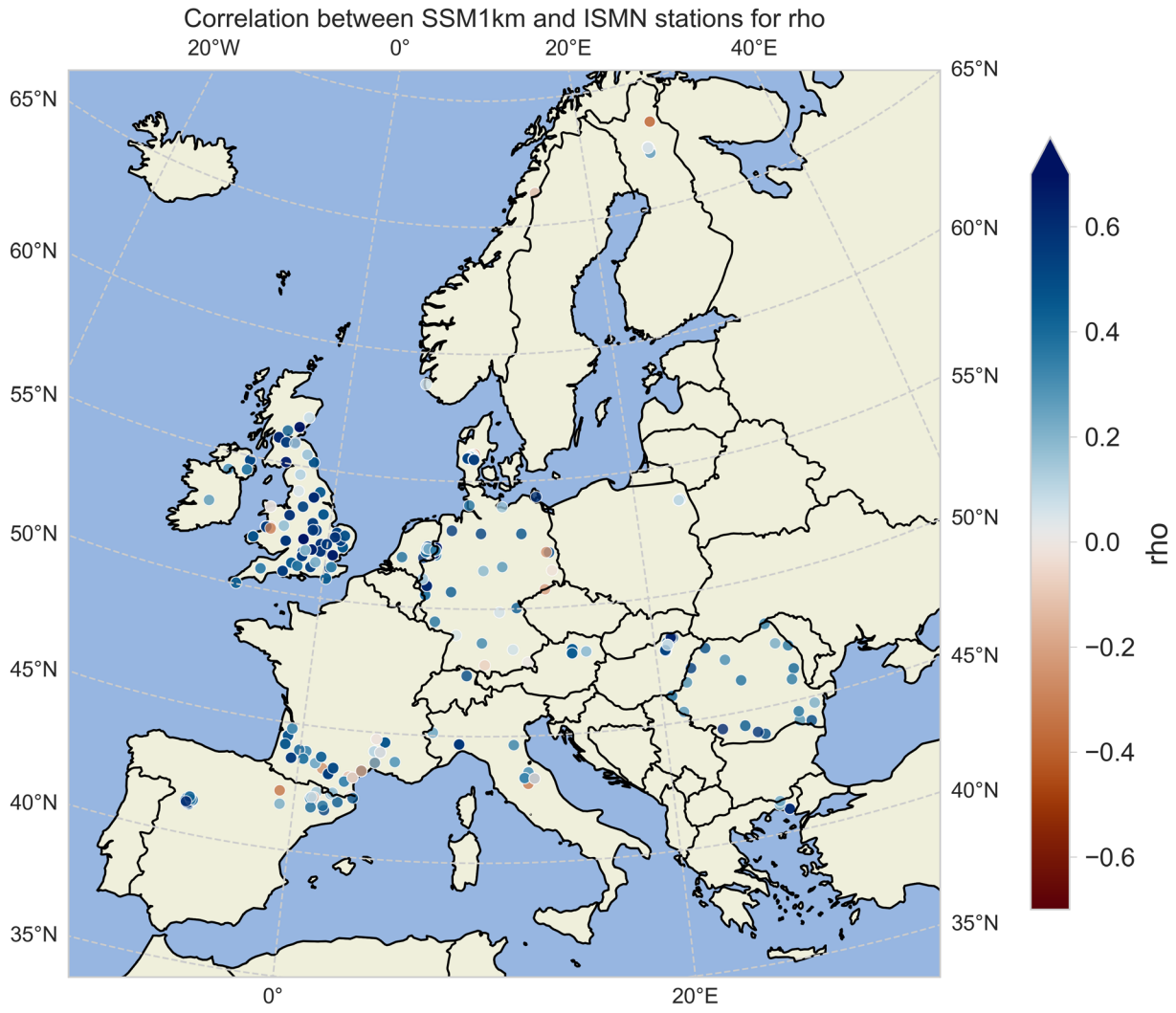


Figure 4: Pearson correlation (rho) over all ISMN stations included in the analysis (period 2015-2024).

A spatial overview of Pearson correlations at all included ISMN stations is shown in Figure 4. As multiple stations overlap in the Europe-wide figure, detailed zoom-in-maps of selected networks are displayed in Figure 5. One can see that the agreement with the SSM1km dataset varies also among stations in the same network. This variation is in some cases similar to what is observed for the SWI1km near surface product (that is SWI with T-value of 2) e.g. for HOAL, BIEBRZA-S-1, TERENO; see [CGLOPS1_VR2024_SWI1km]. Networks with large variation within stations are typically either cover a large surface area (SMOSMANIA, HOBE, GROW, COSMOS-UK), cover different land cover types (HOAL, SMOSMANIA, TWENTE) and/or contain stations with different

temporal availability (HOAL, GROW, COSMOS-UK). All these factors impact the spread of obtained correlation values as observed in Figure 4 and Figure 5. The networks in the images are chosen to showcase the diversity of station recordings in different networks.

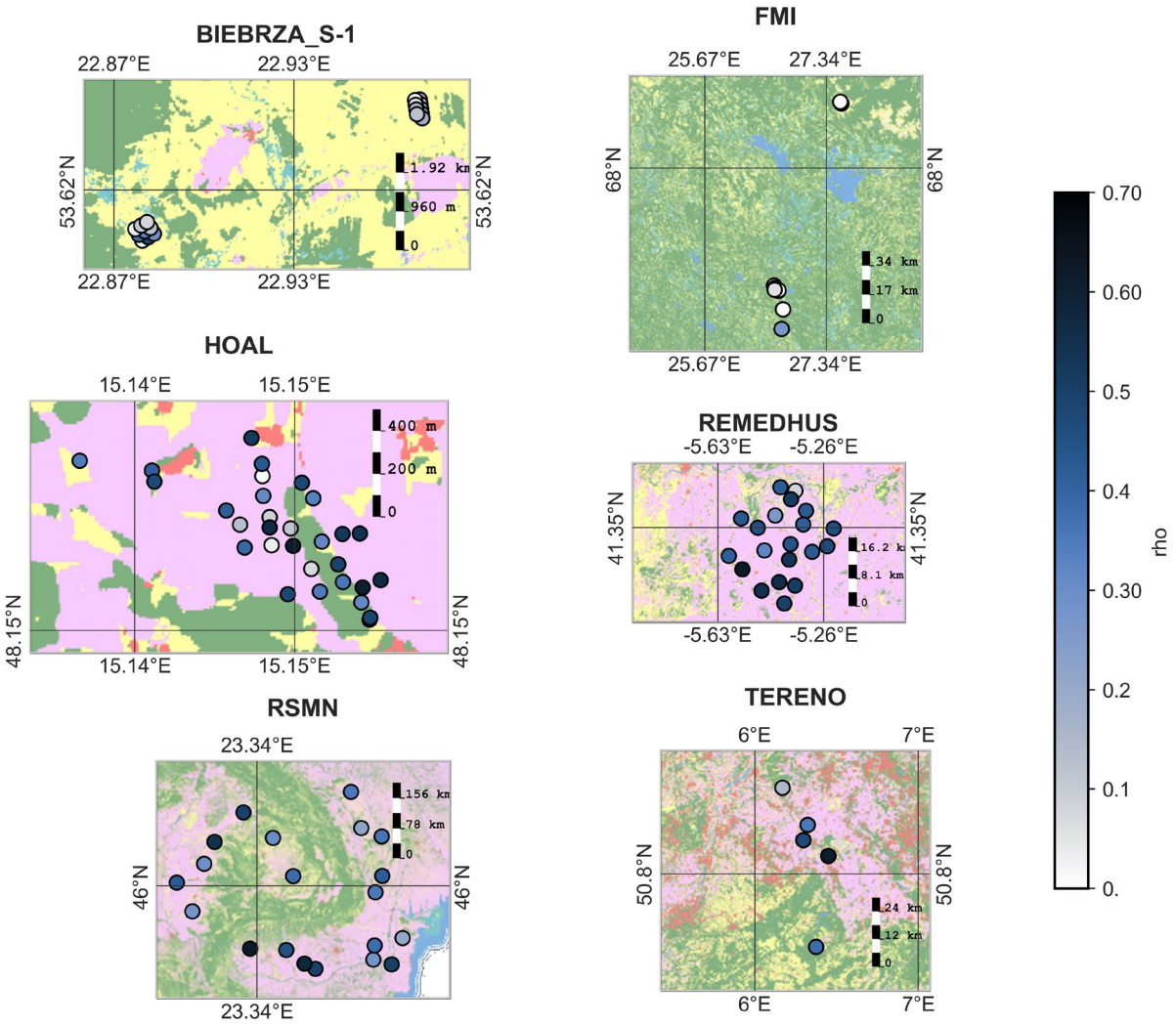
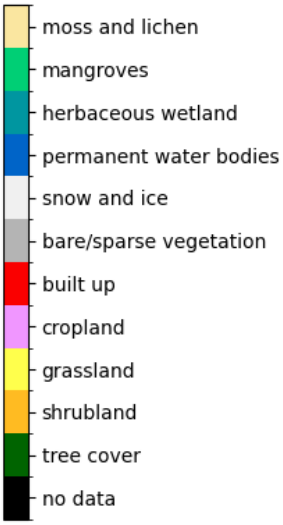


Figure 5: Pearson correlation (rho) over selected ISMN networks. Land cover classes: see Table 3.

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Table 3: Background land cover types for Figure 5 and Figure 9, derived from the ESA WorldCover map 2020.



5.1.2 Summary over Networks: RMSD Analyses

In the following, box plots of RMSD values are given per network, in the same fashion as in the above section on Pearson rho values. The maximum obtained RMSD values range up to 0.26 m^3/m^3, however, the median values range from 0.03 to 0.11$^3/m^3$ (Figure 6).

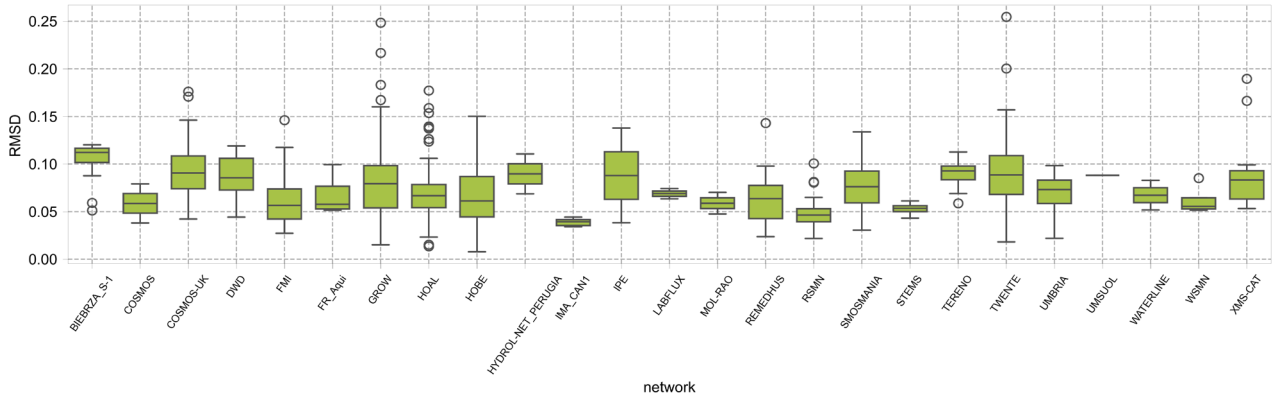


Figure 6: Box plot summarizing RMSD values for SSM calculated over all relevant ISMN stations (0-10 cm depth), per network. Data from 2015-2024 was included. The boxes indicate 25th and 75th percentiles, the whiskers are set to 5th and 95th percentile and outliers are indicated by individual points.

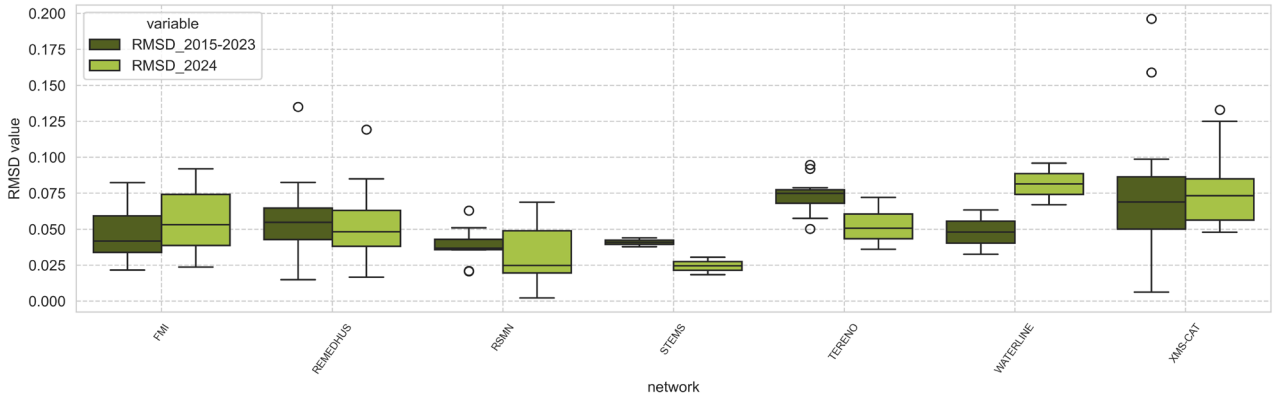


Figure 7: Box plot summarizing RMSD values per network, based on data from 2015-2023 (dark green) and 2024 (light green). Only networks with data in 2024 are included. The boxes indicate 25th and 75th percentiles, the whiskers are set to 5th and 95th percentile, and outliers are indicated by individual points.

The comparison of RMSD values obtained for data from 2024 versus data from 2015-2023 (Figure 7) shows that the RMSD values in 2024 are very similar to the reference, with median RMSD values in 2024 close to 0.05 m³/m³ or below, except for WATERLINE and XMS-CAT. A spatial overview of the RMSD values at all included ISMN stations is shown in Figure 8, and detailed maps of selected networks are shown in Figure 9.

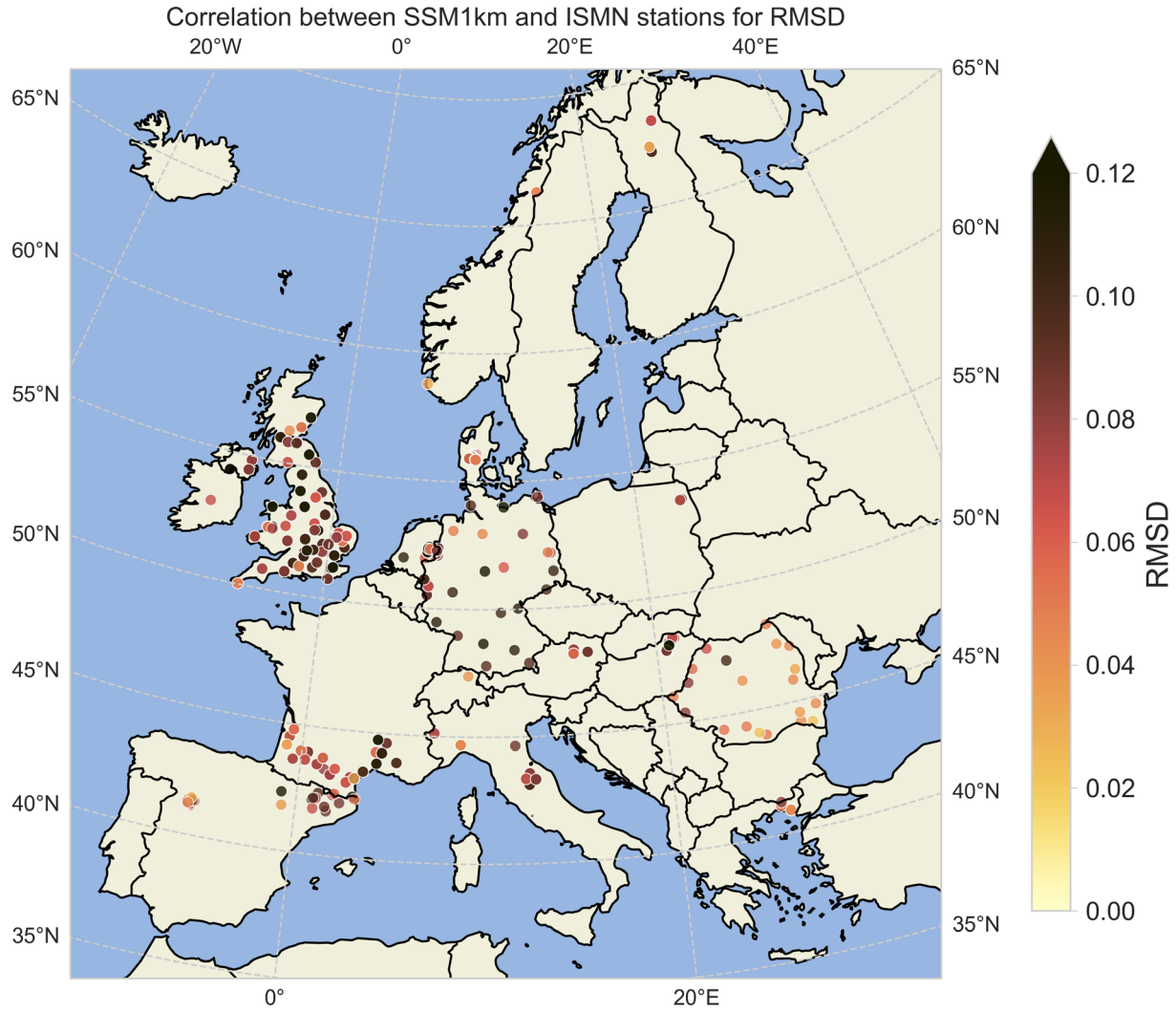


Figure 8: RMSD values over all ISMN stations included in the analysis (period 2015-2024).

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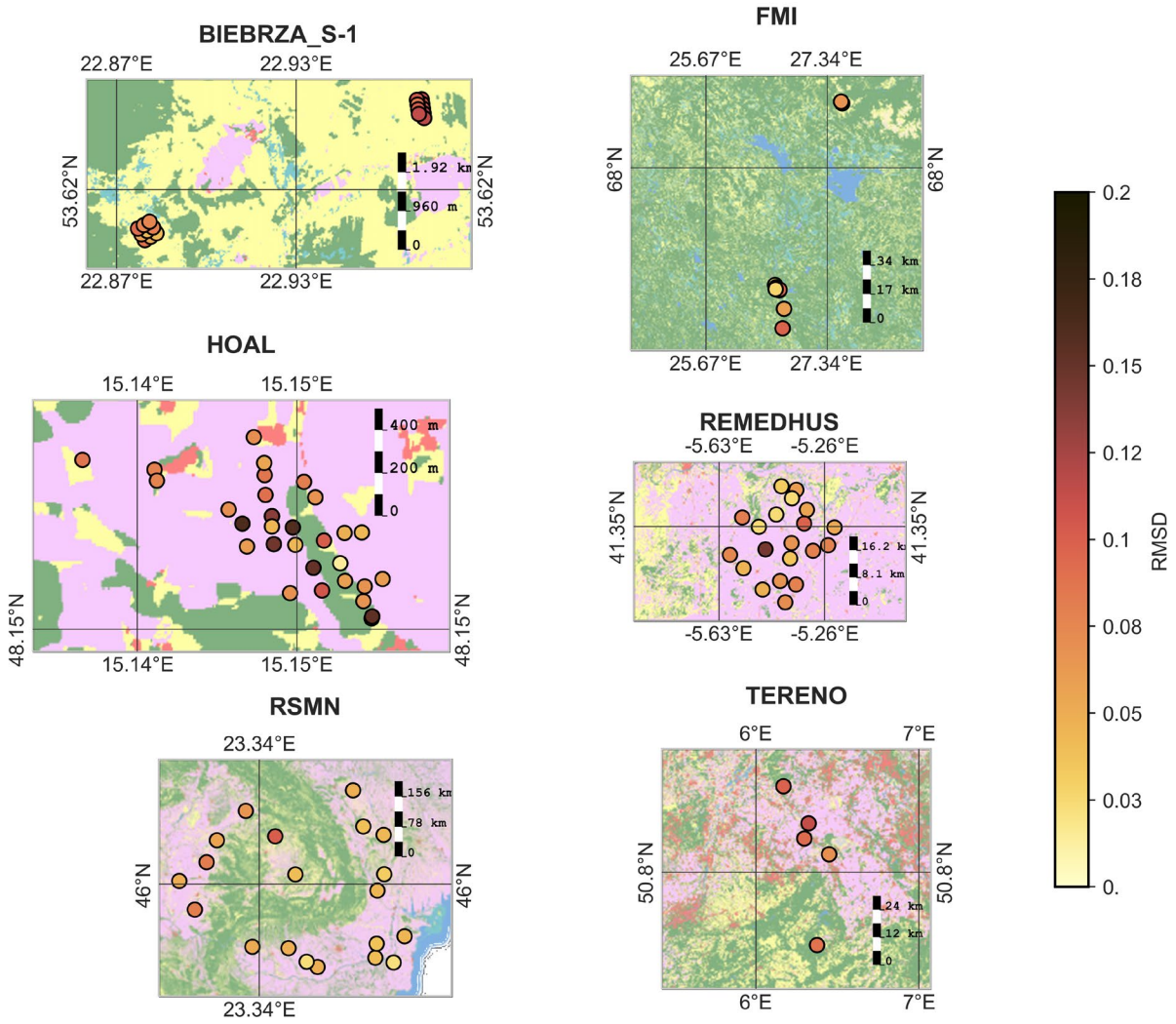


Figure 9: RMSD values over selected ISMN networks. Land cover classes: see Table 3.

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5.2 SELECTED TIME SERIES ANALYSES

In this section, exemplary time series are presented in order to illustrate the general temporal dynamics of satellite and in-situ SM over multiple locations and land cover types. Time series are selected such that they cover a large array of networks across different climate type and soil moisture dynamics in Europe. Each time series includes data for the 2024 period. Except for Figure 16, which exemplifies the COSMOS-UK cosmic ray neutron sensors, kept in this report for its technology that measures soil moisture at a larger footprint (at the scale of hundreds of meters) and thus being closer to the SSM1km product in terms of spatial representativity. Figure 19 highlights a more general challenge with in-situ data, by focusing on multiple stations from STEMS network, where missing or temporally inconsistent data in the year 2024 affects some performance metrics in individual networks, despite the SSM1km product having a satisfactory overall performance.

Strong SM dynamics can be observed in both SSM1km satellite data and in-situ data at many stations. When browsing through Figure 10 to Figure 19, one can see that local peaks and lows are well captured by the SSM1km data, thus reproducing the development of soil moisture changes over the ten years of the evaluation period. From those time series, it is apparent the SSM1km product is especially suitable to reproduce surface soil moisture dynamics in temperate (e.g. RSMN, STEMS, COSMOS-UK, TERENO) to semi-arid climate (e.g. REMEDHUS), as well as humid continental climates (e.g. XMS-CAT), with a vegetation density that is not dampening the SSM signal in the Sentinel-1 SAR data too much. In such environments, the radar signal is also not dominated by complex scattering mechanisms due to vegetation geometry (e.g. dense forests), freezing temperatures, or subsurface scattering during temporally very dry conditions. We note that the XMS-CAT network spans a relatively large area with different climate classifications, with many stations placed in semi-arid regions. For this report, however, we chose to plot the station Los Coscolls (Figure 11) located within a region of humid continental climate.

At some locations from the REMEDHUS, XMS-CAT and TERENO networks (Figure 14 over irrigated cropland; Figure 11 and Figure 12 over cropland - rainfed), distinct seasonal biases during the vegetation period can be observed. This issue was already identified by previous studies on the Sentinel-1 SSM signal quality (e.g. Bauer-Marschallinger et al., 2018) and can be explained by a wrong characterization of vegetation dynamics during the satellite radar backscatter modelling. It can however be seen from the figures that generally a good agreement between satellite and in-situ SM is achieved over multiple stations all across Europe, in particular over cropland and grassland in temperate climate, as measured with a Pearson correlation over 0.6 for the COSMOS-UK station Crichton (Figure 16) and TERENO station Selhausen (Figure 13). Even at arid locations (which are typically challenging for radar-based soil moisture modelling due to non-linear scattering effects from sub-surface scattering during dry conditions, as recently examined by Wagner et al., (2022)), such as

the REMEDHUS-Carretoro station in Spain (with volumetric SM levels below 0.1 throughout the study period), an acceptable Pearson correlation of 0.53 is obtained (Figure 15). While the short-term dynamics are different, the general SM dynamics are captured well by the SSM1km dataset. Stations located in continental climate with cold winters (e.g. over the RSMN network) show less seasonal SSM dynamics but the satellite SSM maintain decent agreement with the in-situ sensors with correlations of 0.36 and 0.50 (Figure 17, Figure 18).

Over forest sites in a cold climate (e.g. over the FMI network, Figure 10), the agreement between SSM1km and in-situ time series is generally lower, as the SSM retrieval in forested areas is challenging due to a number of reasons. First, the sensitivity to soil moisture is lower over densely vegetated areas (Wagner et al., 1999). Second, very distinct, complex scattering mechanisms related to phenological processes such as leaf emergence occur over deciduous broadleaf forest, which complicate the soil moisture estimation. This effect is also observed to a lesser extent for the TERENO station Merzenhausen in Figure 12 (Pfeil et al., 2020, Vreugdenhil et al., 2020). Additionally, the validation over a cold climate at the FMI network is especially challenging as more than 50% of the validation period is masked due to a measured in-situ air/soil temperature below 0°C. A data gap in early summer 2021 (May-July) in the NRT stream of the SSM1km product was reported in 2022, for regions north of 60°N. In the example FMI time series (Figure 10), this data gap thus appears as longer cold season. As a reaction, it is envisaged for future processing chain updates to install dedicated checks on such spatial gaps in the operational NRT stream.

The time series over the STEMS network in Figure 19 compare the measurements of four in-situ stations within the same field and the same SSM pixel (cropland rainfed in subtropical climate). The four in-situ stations have the same characteristics for coordinates, depth, land cover, and sensor brand but with different operational periods. The station with the most observations (Figure 19.d) shows a high agreement with in-situ measurements and a correlation of 0.57. Contrarily, the station, located 50m apart (Figure 19.c) has a correlation of 0.44 and a significantly lower number of in-situ observations, likely due to sparser in-situ SSM availability. When comparing Figure 19.b to 19.d, the number of observations per station within the same analysed field impacts the computed Pearson correlation, as more paired observations yield higher correlation.

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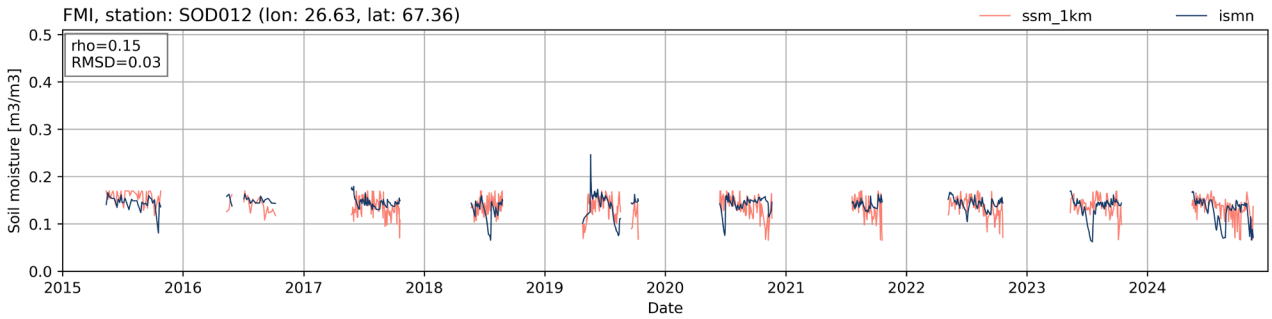


Figure 10: Temporal evolution of in-situ SM (blue) and SSM1km (orange) at FMI station SOD012 at a depth=0.10m (tree cover – needle leaved). The Pearson correlation coefficient (rho) and RMSD value are given in the upper left.

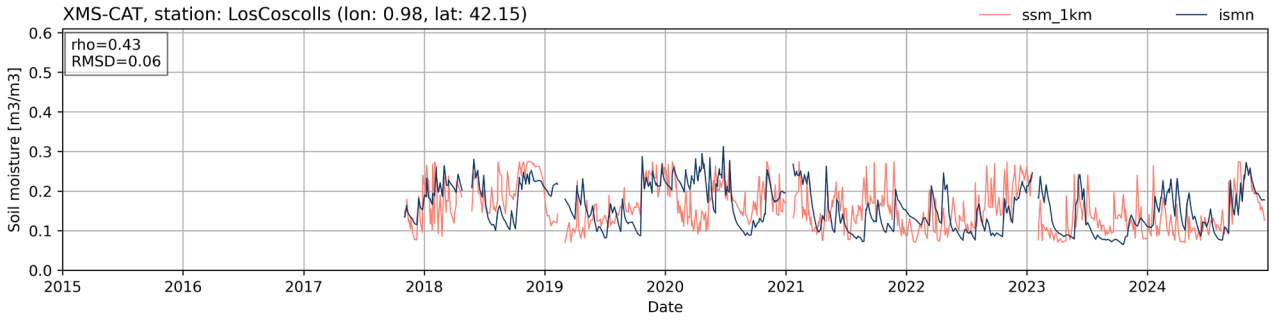


Figure 11: Temporal evolution of in-situ SM (blue) and SSM1km (orange) at XMS-CAT station LosCoscolls at a depth=0.05m (croplands - rainfed).

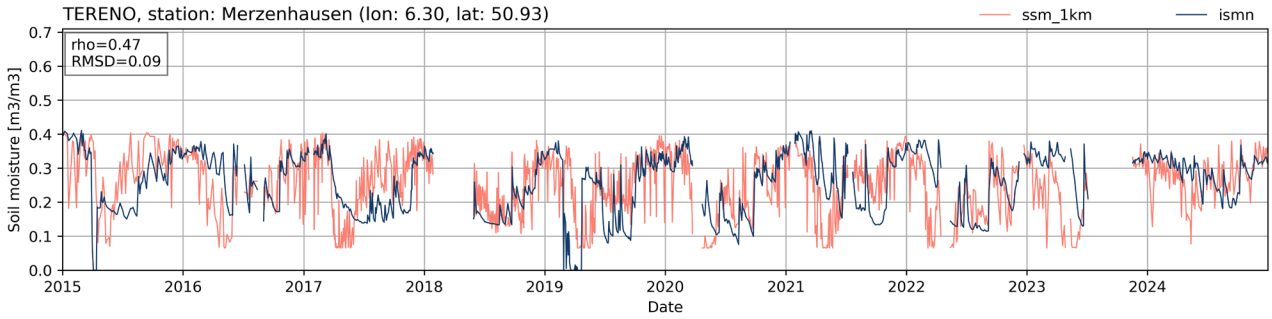


Figure 12: Temporal evolution of in-situ SM (blue) and SSM1km (orange) at TERENO station Merzenhausen at a depth=0.05m (cropland - rainfed).

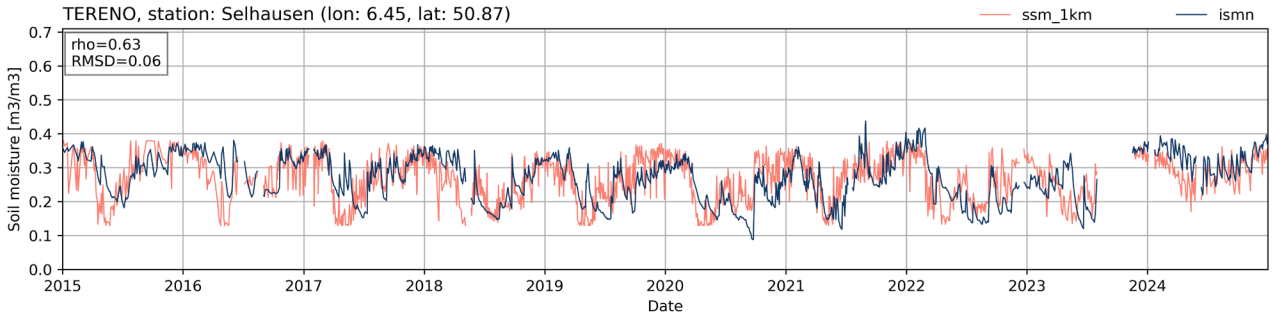


Figure 13: Temporal evolution of in-situ SM (blue) and SSM1km (orange) at TERENO station Selhausen at a depth = [0-0.05]m (cropland - rainfed).

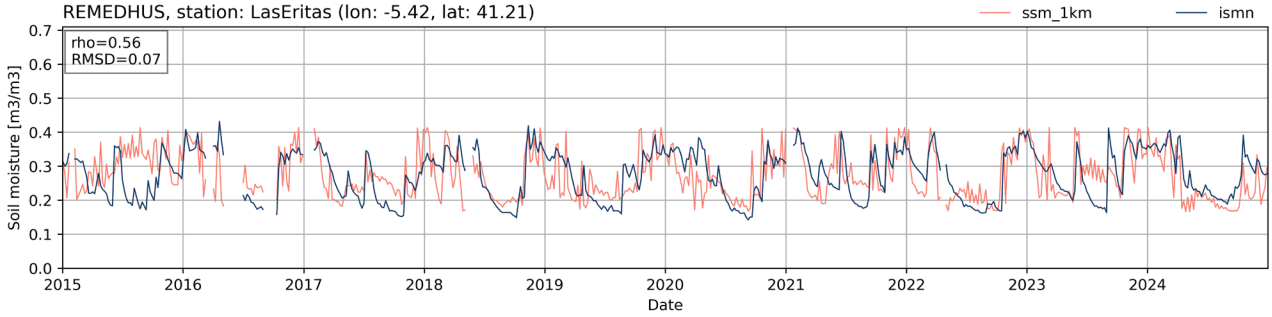


Figure 14: Temporal evolution of in-situ SM (blue) and SSM1km (orange) at REMEDHUS station Las Eritas at depth = [0-0.05]m (cropland - irrigated).

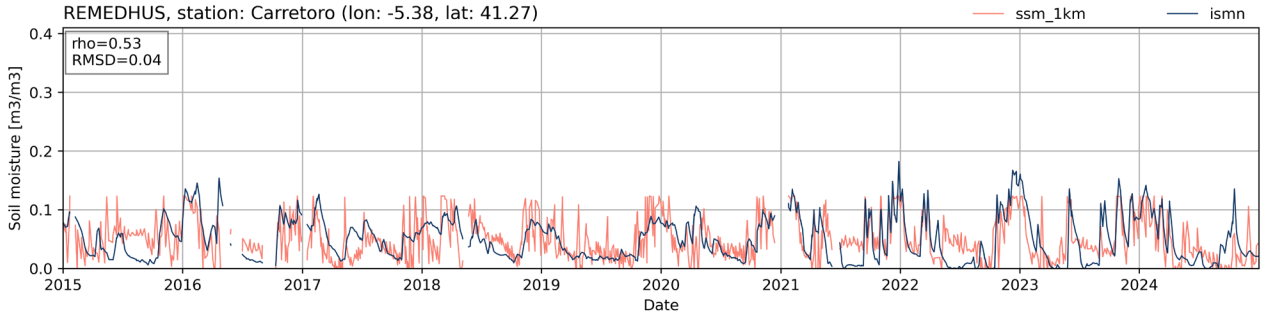


Figure 15: Temporal evolution of in-situ SM (blue) and SSM1km (orange) at REMEDHUS station Carretoro at a depth = [0,0.05]m (cropland - rainfed).

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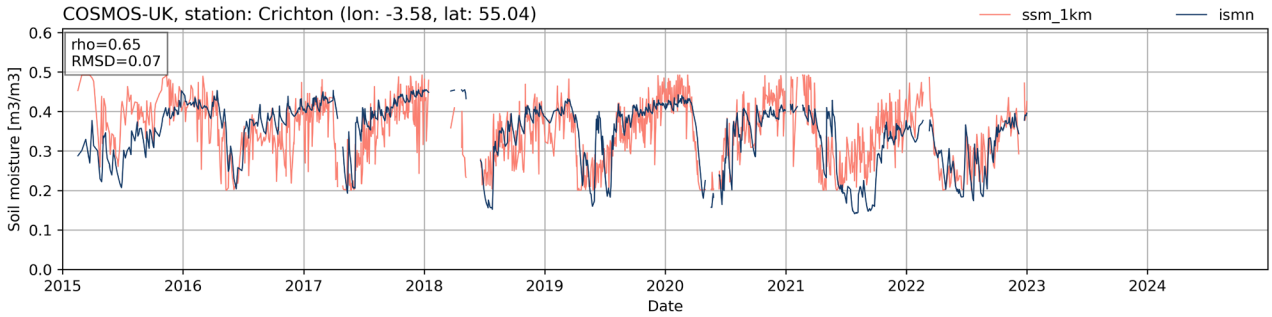


Figure 16: Temporal evolution of in-situ SM (blue) and SSM1km (orange) at COSMOS-UK station Crichton at depth =0.10m (cropland - rainfed).

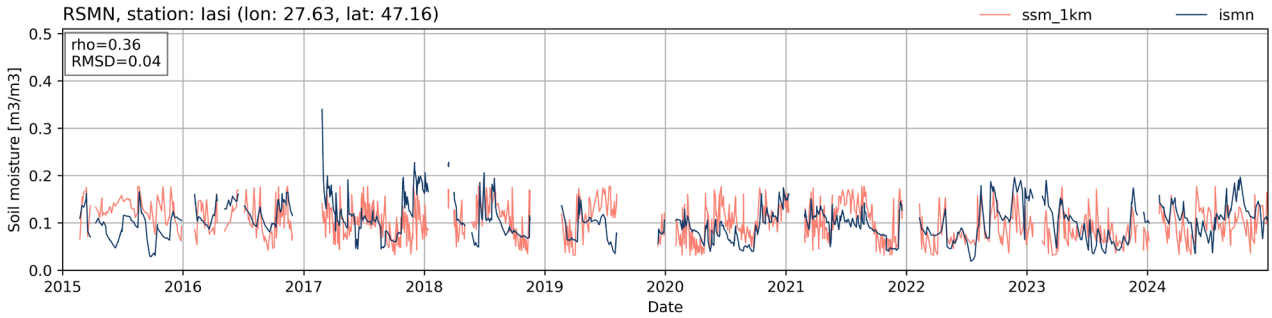


Figure 17: Temporal evolution of in-situ SM (blue) and SSM1km (orange) at RSMN station Iasi at depth = [0-0.05]m (cropland - rainfed).

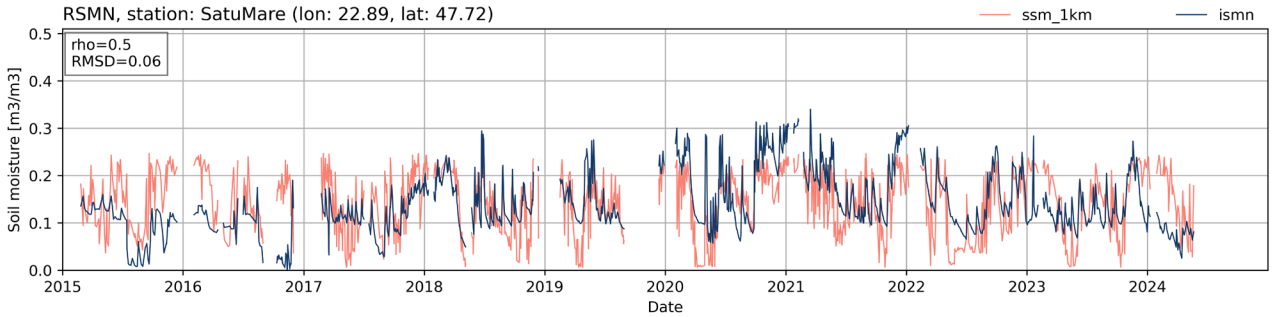
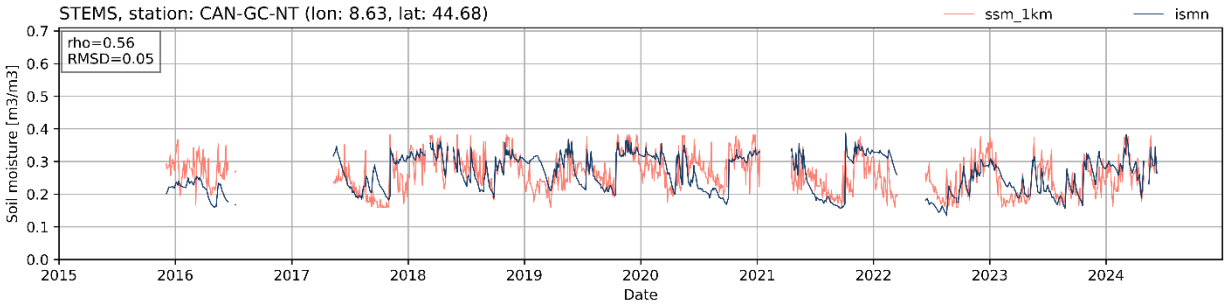
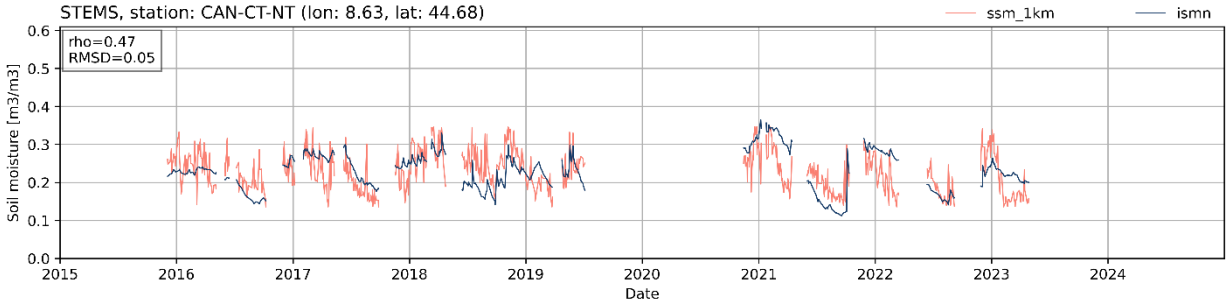


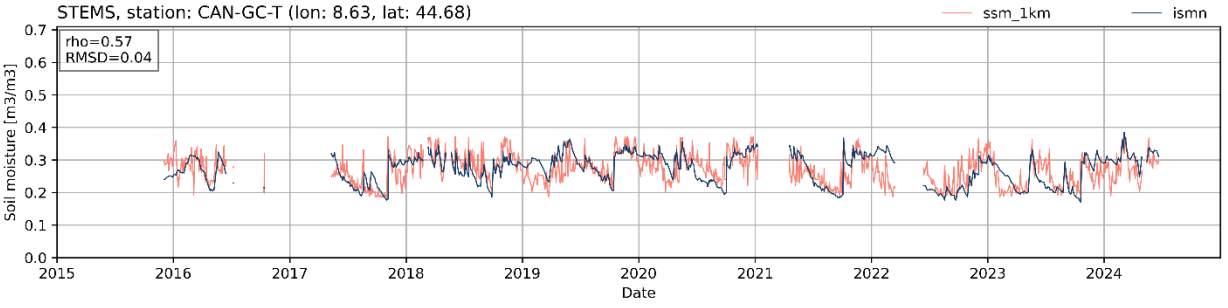
Figure 18: Temporal evolution of in-situ SM (blue) and SSM1km (orange) at RSMN station Satu Mare at depth = [0-0.05]m (cropland - rainfed).



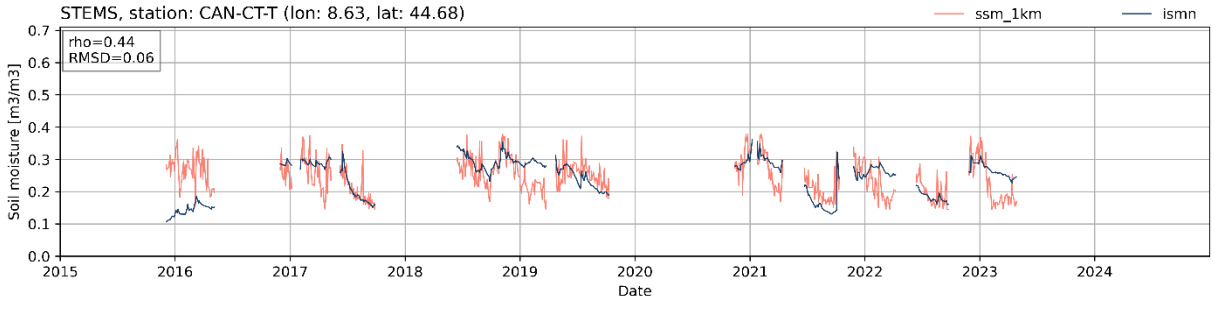
a)



b)



c)



d)

Figure 19: Temporal evolution of in-situ SM (blue) and SSM1km (orange) at STEMS for all four available stations (a to d) at depth = 0.10m over the same rainfed cropland field.

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6 DISCUSSIONS AND CONCLUSIONS

The CLMS SSM1km product provides radar-satellite-based soil moisture estimates from the Sentinel-1 SAR sensor.

This section first highlights the in-depth and peer-reviewed study of Bauer-Marschallinger et al. (2018) over Italy in the period 2015-2017, and then draws the conclusions from the here presented (larger) quality evaluation over Europe in the period 2015-2024.

6.1 SUMMARY FROM DEDICATED PEER-REVIEW ARTICLE

In the peer-reviewed paper, the signal from SSM1km product has been examined for the period 2015-2017 over Umbria. As a 1km model is available for this region, the satellite SSM product was compared to in-situ data as well as the model SM. The analysis yields high agreement over plains and agricultural areas, with low agreement over forests and strong topography. While positive biases during the growing season are detected, the excellent capability to capture small-scale soil moisture changes as from rainfall or irrigation is evident. It was also found that the agreement between SSM1km and the 1km model was better than between SSM1km and in-situ, as the spatial representativeness of the modelled soil moisture values matches the resolution of SSM1km.

6.2 QUALITY ASSESSMENT OVER EUROPE 2015-2024

In this report, the statistical evaluation of the SSM1km product was conducted for the years 2015-2024, with a focus on 2024, using ISMN stations distributed all over Europe. With an average Pearson rho value of 0.39 (weighted by networks number of stations) and an average RMSD of 0.08 m^3/m^3 over all eligible networks, the results are in general moderate. When looking at the scores in more detail, it becomes evident that relatively good scores are achieved over networks in cropland and grassland. Lower correlations on the other hand are mostly found in regions with dense vegetation, wetlands or extreme cold climate. Here, the Sentinel-1 SAR signal is complex and the estimation of SSM is, thus, not straightforward. Also, due to the scale gap between in-situ point measurements and the satellite product's 1km sampling and the consequent representativeness errors, certain deviations are to be expected.

A seasonal bias appears at some stations, showing a partly large under- or overestimation of soil moisture during the growing season. This issue is caused by a misinterpretation of vegetation density change and vitality dynamics in the radar backscatter modelling prior to SSM estimation (Hahn et al., 2017; Pfeil et al., 2018; Bauer-Marschallinger et al., 2018), and are object of currently undergoing method developments with evolutions of the CLMS SM products.

For instance, the current product is known to have reduced sensitivity over densely vegetated areas when validated against both in-situ and modelled data. Research has shown that selectively masking for dense vegetation consistently improves the SSM product sensitivity when validated with ERA5-Land and ISMN stations over Europe. (Massart, et al., 2024).

Apart from the described issues, the time series analysis of the SSM1km dataset revealed the high capability of the SSM1km product to reproduce peaks and lows of the ground soil moisture. The comparison of data from the year 2024 versus data from 2015-2023 showed an overall consistent product quality. For large parts of the sample-set, the 2024 data showed minor differences but general stability. The majority of networks show improvements and the difference of median values of the Pearson correlation between reference and validation period improves by 0.1.

For the 2022-2024 period, a reduced observational frequency due to the Copernicus Sentinel-1B failure leads to a reduction of spatiotemporal coverage of the SSM1km by about 50%. The reduction increases the potential of missing short term hydrological events due to lower available number of observations (like heavy precipitation during thunderstorms). This may result in lower correlation, especially over in-situ stations that show observational gaps too. As seen in Figure 3, the networks displaying a lower correlation are TERENO and WATERLINE, likely due to some specific stations with highly irregular observations through 2024. The remaining networks (FMI, REMEDHUS, RSMN, STEMS, and XMS-CAT) show higher correlation for the year 2024. The successful launch of Sentinel-1C will re-establish a two-satellite constellation (together with Sentinel-1A) and restoration of full mission capacity is expected to be accomplished by mid of 2025.

The overall performance of the SSM1km product is consistent with the previous years' scientific evaluation experiments. The detected differences between the reference period (2015-2023) and the current validation period (2024) are predominantly positive, indicating that SM conditions in 2024 are well represented by the algorithm.

7 RECOMMENDATIONS AND PERSPECTIVES

As the product quality is stable or even improving in the last year, we recommend a continuation of using the SSM1km product. In general, the product features higher data density when having two Sentinel-1 satellite operational, i.e. from 2017-2021 (Sentinel-1A and -1B), and will be again from 2025 onwards (Sentinel-1A and -1C). Users are advised to handle products with timestamps 2015-2016 (lesser density and quality during Sentinel-1A ramp-up) and 2022-2024 (lesser density through roughly halved spatiotemporal coverage due to failure of Sentinel-1B) with more caution, or to apply a more restrictive data selection for those periods.

In particular for land covered by forests, users may be referred to the CLMS SWI1km product as an alternative. Also, over areas with a strong seasonal dynamic in vegetation density, the SWI product could be preferable. However, if the interest is in short-term dynamics of the surface soil layer (e.g. impact from precipitation) the SSM1km provides a strong signal over most land covers.

Current product enhancements are addressing the known issues (vegetation biases, need for masking of frozen surface conditions, floods, topography, and observation geometry effects) and will be implemented through currently ongoing developments within the CLMS evolution activities, leading to an enhanced SSM modelling that includes dynamic vegetation correction, an enhanced masking for subsurface scattering and dense vegetation, and through the introduction of an enhanced frozen surface and observed water body masking, further improving the Surface State Flag (SSF) which is already used for the CLMS global Soil Water Index (SWI) products. Moreover, the Sentinel-1 SAR pre-processing engine currently undergoes technical updates increasing the stability and accuracy of the product. The CLMS users can therefore expect improved quality and reliability of the future SSM1km product version 2.

Within CLMS, preparations to integrate Sentinel-1C observations are well progressed and will recover the SSM1km's coverage. By begin of May 2025, the Sentinel-1C observation scenario is already close to the nominal operations, and the S-1C commissioning activities to complete mission readiness are ongoing and expected to finish mid-2025.

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